

MATH0019 Multivariable Analysis

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0019
<i>Level:</i>	6 (UG), 6 (PG)
<i>Normal student group(s):</i>	UG: Y3 Mathematics programmes PG: MSc Mathematics
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	1
<i>Assessment:</i>	80% examination, 20% coursework
<i>Normal Pre-requisites:</i>	MATH0051 (Analysis 4: Metric Spaces)
<i>Lecturer:</i>	Dr J Hofmann

Course Description and Objectives

This course develops rigorously the important notions and theorems of analysis in $\mathbb{R}^n$. The aim is to unify the classical theorems of vector analysis, applications of which the students have already seen in electromagnetism, with the modern terminology of differential forms.

Recommended Texts

- M Spivak, *Calculus on Manifolds: A Modern Approach to Classical Theorems of Advanced Calculus* (Harper Collins)
- W Rudin, *Principles of Mathematical Analysis* Chapters 9–10.

Detailed Syllabus

- Vector functions, limits and continuity, vector derivatives, total derivative, chain rule, partial derivatives, gradient, inverse function theorem, implicit function theorem.
- Multiple integrals, Fubini theorem, change of variables.
- Differential forms, closed and exact forms, wedge product, simplexes and chains, integration on chains, manifolds, integration on manifolds, Stokes Theorem. The volume element, connections with the classical theorems of vector calculus.